Robert Ondino Morelli

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Education

B.S. Computer Science *[University of Utah]* — *May 2027*

Familiarity

Zig, C/C++, C#, Python, Java, TypeScript/JS, WebAssembly/WasmGC, Qt, Git, Docker, GitHub Actions, Fastlane, CI/CD

Experience

Research Assistant *[University of Utah]* — *Oct 2025 to Present*

- Analyzing performance cost of gradual typing in Meta's Static Python (CinderX)
- Automatic rewriting of Python code to construct test examples
- First author, "Automatic Cast Insertion to Evaluate Progressive Gradual Types" (under review)

Teaching Assistant *[University of Utah]* — *Sep 2025 to May 2026*

- Led labs, graded, and assisted students for COMP 1020

Software Engineer/DevOps *[Stutor Inc.]* — *Sep 2023 to Apr 2024*

- Architected automation pipeline using Fastlane and GitHub Actions to build, sign and deploy to multiple platforms
- Added compound keys to tables where backend queries had drifted from database schema; query times reduced from 15+ s to <1 s

Web Developer/DevOps *[Jerran Software Solutions]* — *Apr 2022 to Sep 2023*

- Software consulting; clients included the LDS Church, Retro Game Vault and Stutor Inc.
- Architected, proposed and implemented overhaul of development process for the LDS MTC team using build automation in conjunction with new QA practices; deployment was reduced from a week-long headache to mere minutes
- Rewrote Embark startup using dependency-ordered queue for async initializations; first-time loading reduced by 50%

Research Assistant Intern *[Earl Keefe PhD, University of Utah]* — *Nov 2020 to Jun 2021*

- Used pypng to construct novel visualizations of covariance matrices to provide more intuitive understanding of data
- Implemented statistical formulas in clear code in order that a reviewer can be sure the results are correct and valid

Web Development Intern *[Frelii]* — *Jun 2019 to Sep 2019*

- Implemented web scraping SNPedia for AI training using Python with multiprocessing and BeautifulSoup

Projects

IRX regex fuzzer *[Zig]* ↗

- The inverted regex fuzzer generates accept and reject string seeds and fuzzes patterns instead of inputs across ~25 engines

UTU language and compiler *[JS/TS, WasmGC]* ↗

- The language AssemblyScript was meant to be: designed from the ground up to have predictable lowering to WasmGC, have small bundle size, and support .d.ts as an ABI

Spreadsheet formulas to DLL compiler *[C#]* ↗

- Compiles spreadsheet formulas into a DLL, supporting optimizations including fast-math style unsafe float conversion

Color alchemy *[Qt/C++]* ↗

- Beautiful game for learning color mixing, properly using OKLab color space for gradient construction and scoring

Open source & competitions

Contributed to open-source IDE code-base *[MARS]* ↗

- Discovered and corrected oversight in the MARS IDE's instruction decoder that affected correctness of self-modifying code

CinderX bug hunting *[CinderX]*

- Reported 11 bugs in CinderX and contributed 2 fixes to the type-checker

Regex bug hunting

- Reported 46 correctness bugs across a wide variety of regular expression engines including Perl, .NET and V8

Competitive programming *[ICPC]* ↗

- As part of team "Big Cottonwood Coders", placed 12 out of 48 teams at the Rocky Mountain regional contest (2025)
- As Event Coordinator of the U of U technical coding club, organized and oversaw competitions (2026)

Jane Street leaderboard

- Number Cross 5 ↗
- Sum One, Somewhere ↗